Computational Notebooks in Python, R, and Stata

A Multi-Language Guide to Data Science Workflows

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This document introduces computational notebooks as a practical tool for data science across three programming languages: Python, R, and Stata. Through five companion notebooks, we demonstrate how Jupyter-based environments support reproducible workflows ranging from exploratory data analysis to machine learning and causal inference. The first three notebooks present a common data exploration task implemented independently in Python, R, and Stata, illustrating how each language handles data generation, visualization, and summary statistics within the same notebook platform. The fourth notebook introduces Random Forest regression for predicting Bolivia's Municipal Sustainable Development Index from satellite image embeddings. The fifth notebook demonstrates Double Machine Learning for causal inference using the Pennsylvania Reemployment Bonus experiment. Together, these notebooks provide a hands-on reference for researchers and students working across multiple computational environments.

1 Introduction

Computational notebooks have become a standard tool for reproducible data science, combining code, output, and narrative in a single document. While Python dominates the Jupyter ecosystem, researchers in the social sciences and economics often work in R or Stata. This document demonstrates that all three languages can be used effectively within Jupyter notebooks, and that the same analytical tasks can be accomplished across environments.

The companion notebooks use data from the DS4Bolivia project as a running example. The dataset covers 339 Bolivian municipalities and includes the Municipal Sustainable Development Index (IMDS), 64-dimensional satellite image embeddings, and regional identifiers.

The five notebooks are organized as follows:

- N1: Sample Data Exploration — Exploratory data analysis in Python
- N2: Sample R Analysis — The same exploration in R
- N3: Sample Stata Analysis — The same exploration in Stata
- N4: Introduction to Machine Learning — Random Forest Regression — Predicting sustainable development from satellite embeddings
- N5: Introduction to Causal Inference — Double Machine Learning — Estimating causal effects with debiased ML methods

Each section below introduces one notebook, summarizes its content, and embeds selected figures and tables. Readers are encouraged to open the full notebooks for complete code and discussion.

2 Data Exploration in Python

The first notebook (N1: Sample Data Exploration) demonstrates a standard exploratory data analysis workflow in Python. It generates a synthetic dataset of 80 regions grouped into four geographic clusters, each with simulated GDP per capita and life expectancy values. The notebook illustrates how to set up a reproducible environment using a shared configuration file (`config.py`), create publication-quality visualizations with Matplotlib and Seaborn, and compute summary statistics with Pandas.

The scatter plot below shows the relationship between GDP per capita and life expectancy across the simulated regions, colored by geographic cluster.

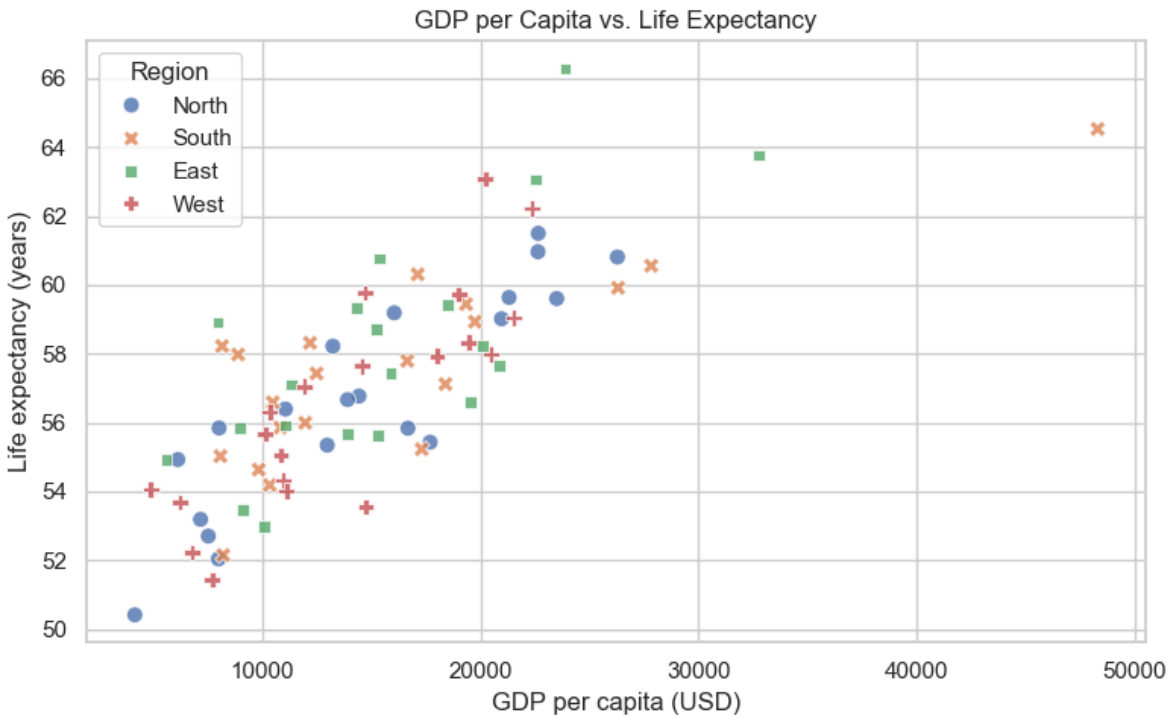

Figure 1: Synthetic regional indicators: GDP per capita vs. life expectancy across 80 simulated regions, colored by geographic cluster.

The following table reports summary statistics by region.

Table 1: Summary statistics by region.

Region	GDP per capita (USD) (mean)	GDP per capita (USD) (std)	GDP per capita (USD) (count)	Life expectancy (years) (mean)	Life expectancy (years) (std)	Life expectancy (years) (count)
East	15607.9	6499.5	20	58.1	3.4	20
North	14709	6638.8	20	56.7	3.1	20
South	16113.3	9541.1	20	57.5	2.8	20
West	13787	5498.2	20	56.7	3.2	20

This notebook serves as a template that researchers can adapt to their own datasets. The key takeaway is the basic workflow: load data, visualize distributions and relationships, and summarize key variables — all within a single reproducible document.

3 Data Exploration in R

The second notebook (N2: Sample R Analysis) replicates the same exploratory analysis using R. It demonstrates how to run R code inside a Jupyter notebook using the IRkernel, generate equivalent visualizations with ggplot2, and produce formatted tables with `knitr::kable()`.

The scatter plot below shows the same GDP-life expectancy relationship rendered with ggplot2.

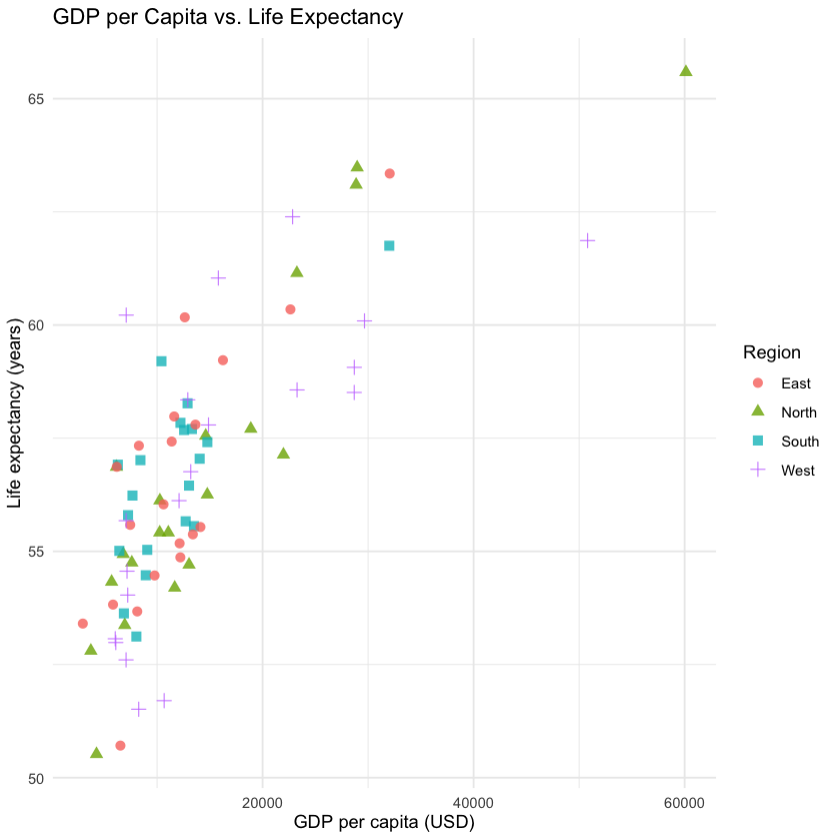

Figure 2: Synthetic regional indicators: GDP per capita vs. life expectancy across 80 simulated regions, colored by geographic cluster (R).

Although this notebook is straightforward, it highlights several practical points for R users: configuring an R kernel in Jupyter, sourcing a shared configuration file (`config.R`) for reproducibility, and using R's visualization and table-formatting ecosystem within the notebook environment. For researchers more comfortable in R than Python, this notebook provides a familiar entry point into the Jupyter workflow.

Table 2: Summary statistics by region (R).

```
|Region | GDP mean|  GDP sd| Life exp. mean| Life exp. sd|  N|
|:------|--------:|-------:|--------------:|------------:|--:|
|East   |  11888.0|  6419.8|           56.5|          2.9| 20|
|North  |  15443.4| 13021.0|           56.8|          3.8| 20|
|South  |  11517.9|  5595.0|           56.6|          2.0| 20|
|West   |  15972.3| 11548.6|           56.8|          3.5| 20|
```

4 Data Exploration in Stata

The third notebook (N3: Sample Stata Analysis) completes the multi-language comparison by performing the same analysis in Stata. It uses the `nbstata` kernel, which allows Stata commands to run directly inside Jupyter notebooks — a relatively recent development that brings Stata into the computational notebook ecosystem.

The scatter plot below shows the GDP-life expectancy relationship using Stata's `twoway scatter` graphics.

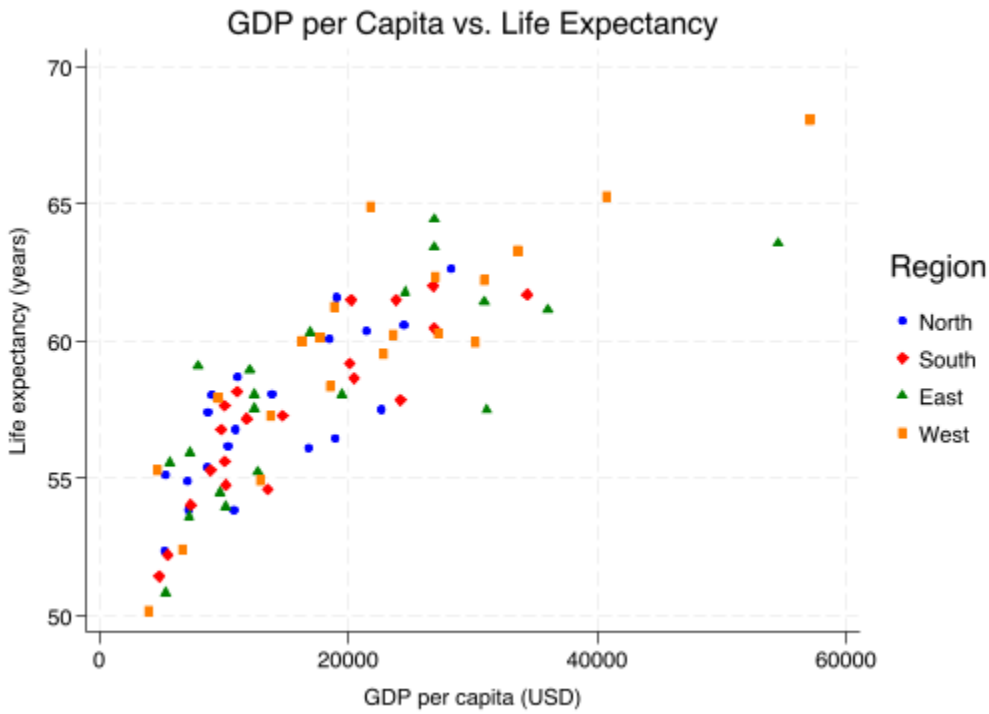

Figure 3: Synthetic regional indicators: GDP per capita vs. life expectancy across 80 simulated regions, colored by geographic cluster (Stata).

```
Summary statistics: Mean, SD, N
Group variable: region

region |  gdp_pe~a  life_e~y
-------+--------------------
  East |   18540.3      58.2
       |   12777.3       3.7
       |      20.0      20.0
-------+--------------------
 North |   13941.3      57.3
       |    6845.2       2.8
       |      20.0      20.0
-------+--------------------
 South |   15757.6      57.4
       |    8297.8       3.1
```

```
       |      20.0      20.0
-------+--------------------
  West |   21923.5      59.7
       |   12889.5       4.3
       |      20.0      20.0
-------+--------------------
 Total |   17540.7      58.2
       |   10782.0       3.6
       |      80.0      80.0
----------------------------
```

This notebook is particularly valuable for researchers who rely on Stata for their primary analysis. It demonstrates that Stata users do not need to abandon their preferred tool to benefit from the reproducibility and documentation advantages of computational notebooks. The notebook covers data generation with Stata syntax, graphing with `twoway`, and summary statistics with `tabstat`.

5 Machine Learning with Random Forest

The fourth notebook (N4: Introduction to Machine Learning — Random Forest Regression) moves beyond exploratory analysis to introduce a complete machine learning workflow in Python. It predicts Bolivia's Municipal Sustainable Development Index (IMDS) from 64-dimensional satellite image embeddings using Random Forest regression.

5.1 Target variable

The IMDS scores across 339 municipalities range from 0 to 100 and follow a roughly bell-shaped distribution with a slight right skew (mean = 51.1, median = 50.5).

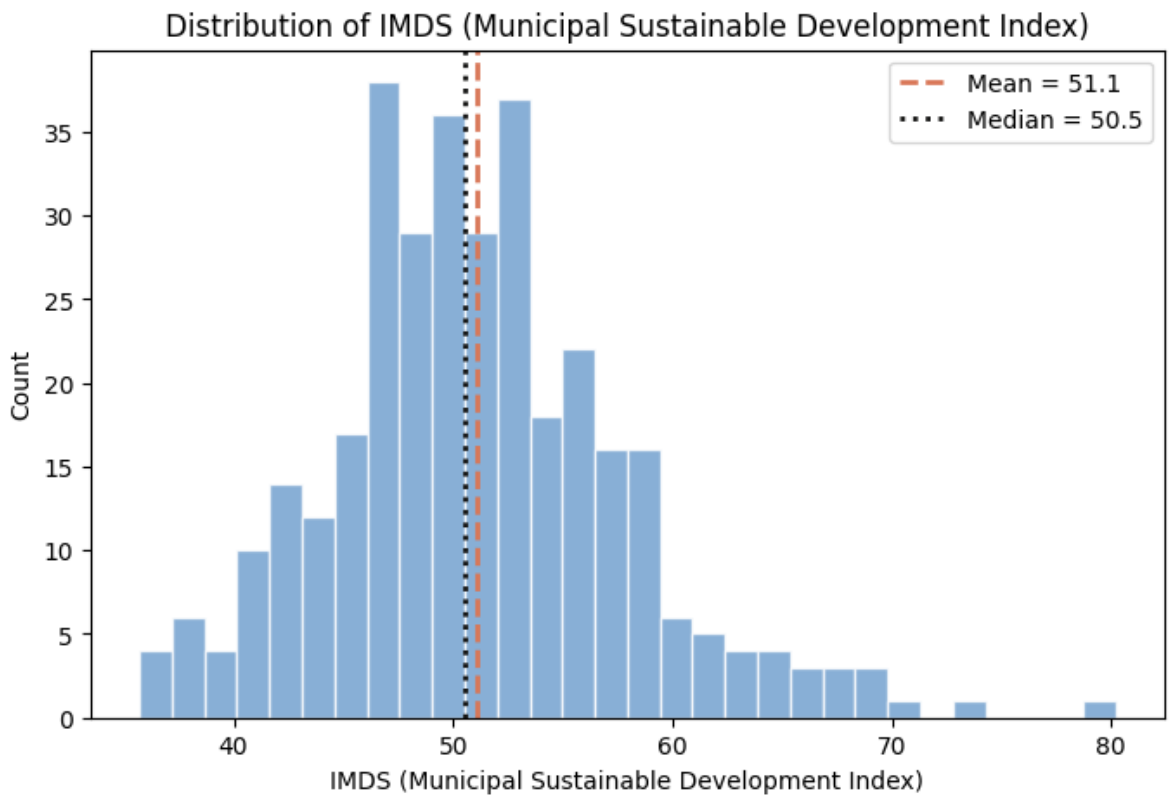

Figure 4: Distribution of IMDS scores across Bolivia's municipalities. The dashed line marks the mean, the dotted line marks the median.

5.2 Model performance

The dataset is split into 271 training and 68 test municipalities. A baseline Random Forest model is trained with 5-fold cross-validation, followed by hyperparameter tuning via randomized search over 50 parameter combinations. The table below compares the baseline and tuned models on the held-out test set.

Table 3: Random Forest regression results: baseline vs tuned model performance on the test set.

Metric	Baseline	Tuned
R^2	0.2307	0.2297
RMSE	6.52	6.52
MAE	4.68	4.72

Both models achieve an R-squared of approximately 0.23 and an RMSE of about 6.5 points, indicating that satellite embeddings capture meaningful but limited development signals. The scatter plot of actual versus predicted values illustrates the typical regression-to-the-mean pattern.

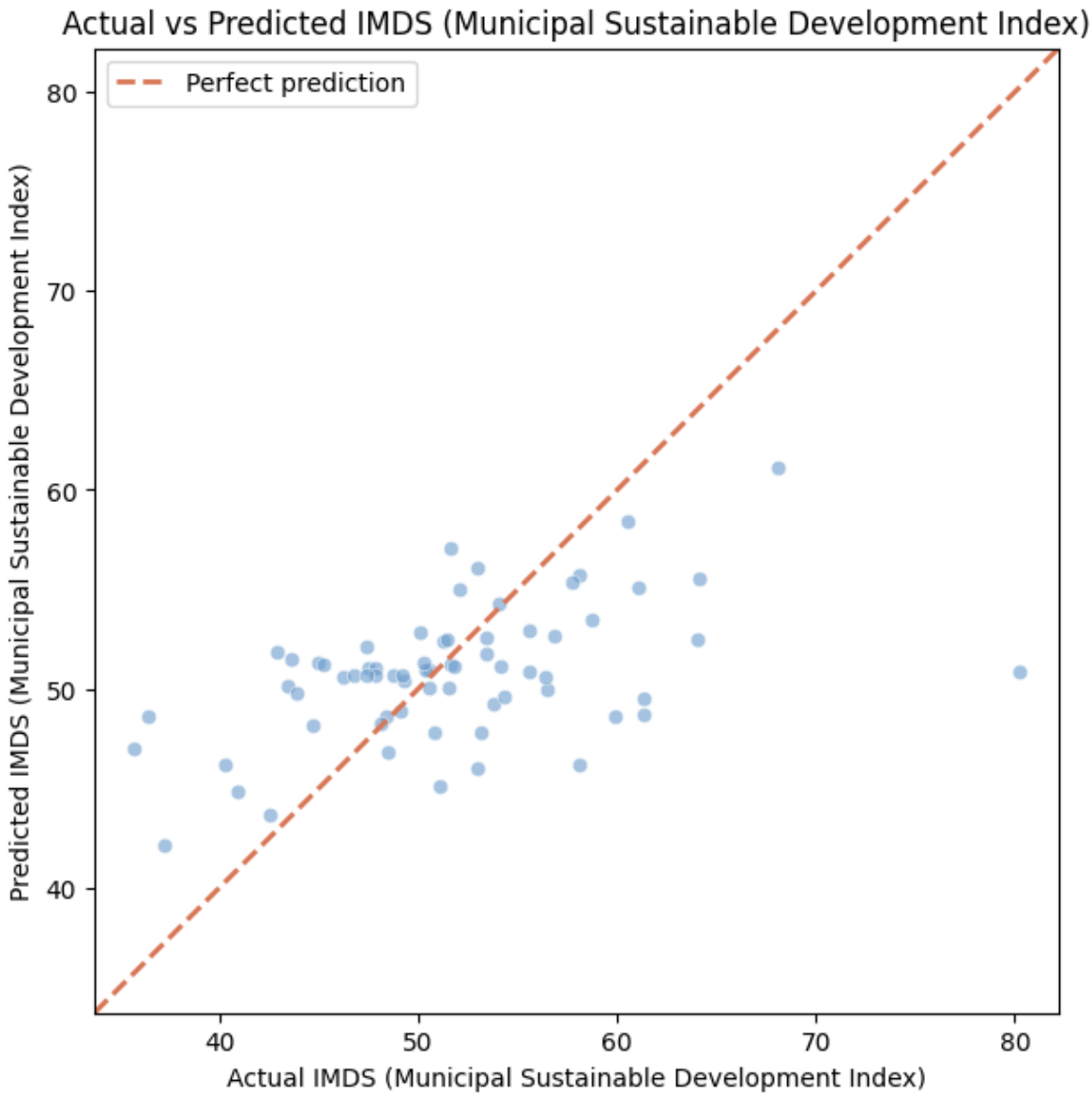

Figure 5: Actual vs predicted IMDS scores on the test set. The dashed line represents perfect prediction.

5.3 Feature importance

A key advantage of Random Forest models is their interpretability through feature importance measures. The permutation importance plot below shows the top features ranked by their contribution to predictive accuracy.

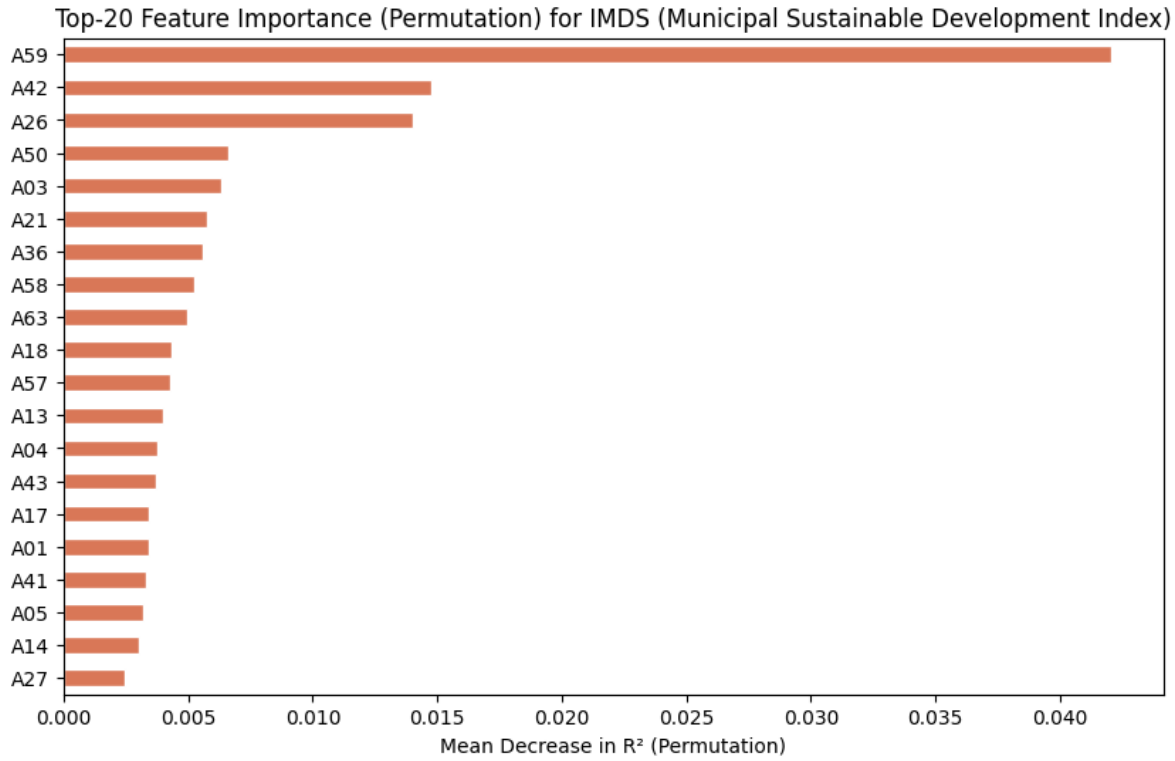

Figure 6: Top-20 satellite embedding features ranked by permutation importance (mean decrease in R^2 when feature is shuffled).

Importance is distributed across many embedding dimensions rather than concentrated in a few, suggesting that multiple aspects of the satellite imagery contribute to predicting sustainable development outcomes.

5.4 Partial dependence

Partial dependence plots reveal non-linear relationships between individual features and the predicted IMDS. Several features exhibit threshold effects where the marginal relationship changes at specific values.

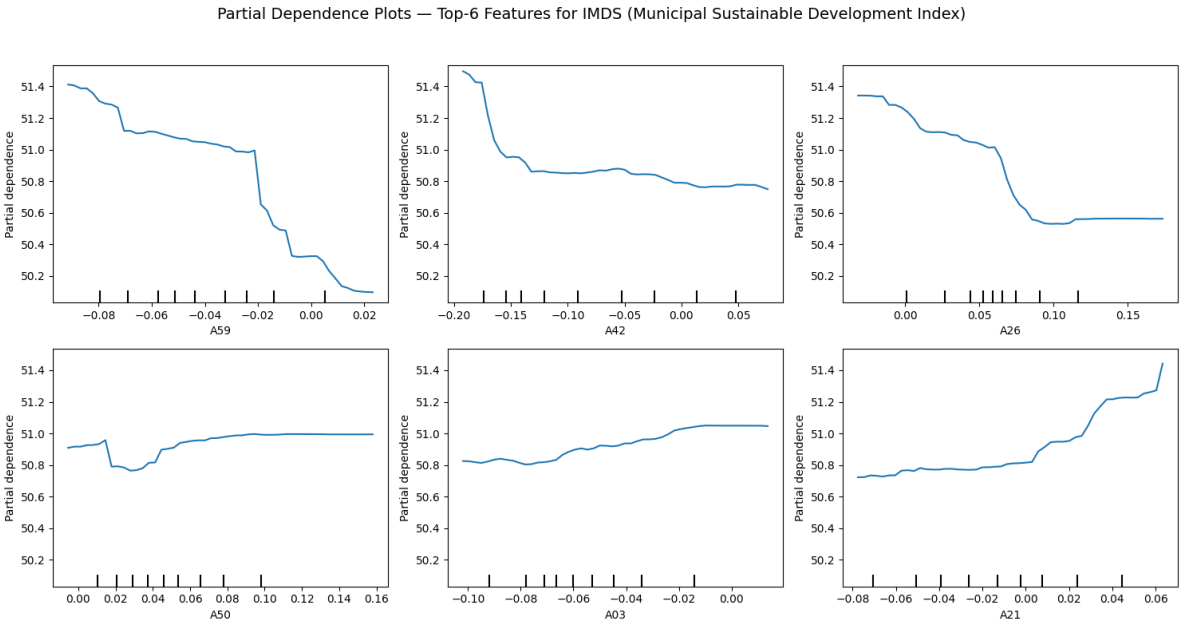

Figure 7: Partial dependence plots for the top-6 most important satellite embedding features, showing how each feature's value affects the predicted IMDS score.

For full details on the methodology and hyperparameter search, see the complete notebook.

6 Causal Inference with Double Machine Learning

The fifth notebook (N5: Introduction to Causal Inference — Double Machine Learning) introduces causal inference methods using the Pennsylvania Reemployment Bonus experiment as a case study. It demonstrates how Double Machine Learning (DoubleML) corrects for the confounding bias that affects naive regression estimates.

6.1 The research question

The central question is whether offering a cash bonus to unemployed workers causally reduces the duration of unemployment. The dataset includes 5,099 workers — 3,354 in the control group and 1,745 who received the bonus treatment.

Figure 8: Distribution of log unemployment duration by treatment group. The bonus group shows a slightly lower mean.

6.2 Comparing estimation methods

The notebook compares four estimation approaches: naive OLS (without covariates), OLS with covariates, DoubleML with Random Forest learners, and DoubleML with Lasso learners. Naive OLS overestimates the treatment effect by approximately 17% compared to the debiased DoubleML estimates.

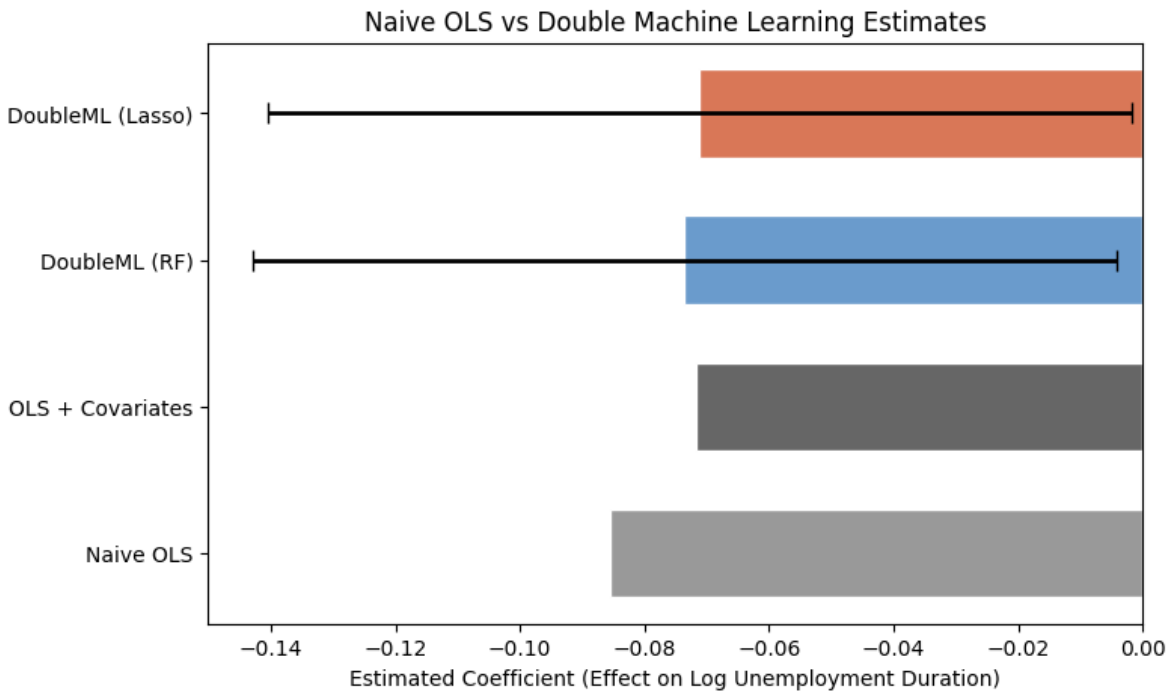

Figure 9: Comparison of causal effect estimates across methods. Error bars show 95% confidence intervals for DoubleML methods.

6.3 Results

The DoubleML estimates indicate that the bonus reduces log unemployment duration by approximately 0.07 (about 7%), and this effect is statistically significant. The table below summarizes the results across all methods.

Table 4: Comparison of treatment effect estimates across methods.

Method	Coefficient	Std Error	p-value	95% CI
Naive OLS	-0.0855	—	—	—
OLS + Covariates	-0.0717	—	—	—
DoubleML (RF)	-0.0736	0.0354	0.0378	[-0.1430, -0.0041]
DoubleML (Lasso)	-0.0712	0.0354	0.0445	[-0.1406, -0.0018]

The confidence intervals for the DoubleML estimates lie entirely to the left of zero, confirming statistical significance. The results are robust across different ML learners (Random Forest and Lasso).

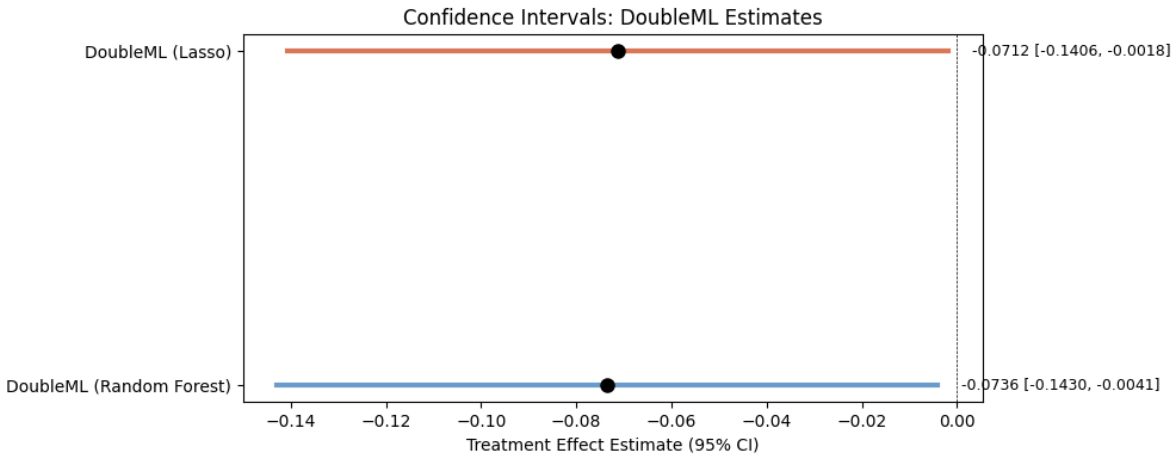

Figure 10: 95% confidence intervals for the DoubleML causal effect estimates using two different ML learners.

For the full methodological treatment — including the Partially Linear Regression framework, cross-fitting procedure, and discussion of assumptions — see the complete notebook.

7 Concluding Remarks

This document has introduced five computational notebooks spanning three programming languages and three analytical approaches:

- **Notebooks 1–3** demonstrate that Python, R, and Stata can all operate within the Jupyter notebook environment, producing comparable results for the same exploratory analysis task. This lowers barriers for researchers who prefer one language but want the benefits of reproducible, self-documenting workflows.
- **Notebook 4** provides a hands-on introduction to machine learning with Random Forest regression, showing how satellite image embeddings can predict municipal development outcomes — explaining approximately 23% of the variance in Bolivia's IMDS.
- **Notebook 5** introduces causal inference with Double Machine Learning, illustrating how debiased estimation corrects for confounding bias relative to naive OLS approaches.

Together, these notebooks serve as a practical reference for researchers and students learning to work with computational notebooks across multiple programming environments.

8 Acknowledgments

This research uses data from the DS4Bolivia project. The analytical workflow was developed at Nagoya University.

9 Data and Code Availability

The data and code used in this study are available at https://github.com/quarcs-lab/ds4bolivia. All notebooks are fully reproducible using the `uv` environment manager and the dependencies specified in `pyproject.toml`.

References